TechInovasi Stack

Professional IT Infrastructure Automation with Docker

Document Version: 1.0

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Status: Production Ready

Server IP: 10.10.10.1

Base OS: Debian 12 Bookworm

BAB 1: PENDAHULUAN & ARSITEKTUR

1.1 Latar Belakang

TechInovasi Stack merupakan solusi infrastruktur teknologi informasi terintegrasi yang dikembangkan untuk memenuhi kebutuhan perusahaan di bidang teknologi. Solusi ini mengimplementasikan otomatisasi penuh terhadap proses instalasi dan konfigurasi layanan-layanan kritis yang secara tradisional dikelola melalui prosedur manual. Pendekatan ini secara signifikan mengurangi potensi kesalahan manusia (human error) serta mengoptimalkan efisiensi waktu operasional tim teknis.

1.2 Tujuan Implementasi

Tujuan	Deskripsi
Automasi Total	Eksekusi satu perintah (docker compose up -d) untuk mengaktifkan seluruh layanan secara simultan
Keamanan Terintegrasi	Penerapan FTPS (FTP over TLS) untuk transfer file terenkripsi dan SSH key-based authentication untuk akses remote
Persistensi Data	Seluruh data database, file konfigurasi, dan dokumen pengguna tetap terjaga meskipun container mengalami restart atau rebuild
Portabilitas	Kemampuan untuk dipindahkan ke lingkungan server lain tanpa memerlukan konfigurasi ulang yang ekstensif

1.3 Arsitektur Sistem

Berikut adalah gambaran arsitektur sistem secara keseluruhan:

text

```
┌──────────────────────────────────────────────────────────────────┐
│                          DOCKER HOST                             │
│                          (10.10.10.1)                            │
├──────────────────────────────────────────────────────────────────┤
│   ┌──────────────┐   ┌──────────────┐   ┌──────────────┐        │
│   │  Company     │   │  WordPress   │   │   MariaDB    │        │
│   │  Profile     │   │   Apache     │   │   Database   │        │
│   │   Apache     │   │   :8081      │   │  (internal)  │        │
│   │   :8080      │   └──────┬───────┘   └──────┬───────┘        │
│   └──────────────┘          │                  │                │
│                             └──────────┬───────┘                │
│                                        │                        │
│   ┌──────────────┐   ┌──────────────┐ │                        │
│   │    FTPS      │   │     SSH      │ │                        │
│   │   vsftpd     │   │  OpenSSH     │ │                        │
│   │    :21       │   │   :2222      │ │                        │
│   └──────────────┘   └──────────────┘ │                        │
│                                        │                        │
│   ┌────────────────────────────────────┴──────────────────────┐ │
│   │                   DOCKER NETWORK (internal)               │ │
│   └───────────────────────────────────────────────────────────┘ │
└──────────────────────────────────────────────────────────────────┘
                                  │
                   ┌──────────────┴──────────────┐
                   ▼                             ▼
          ┌──────────┐   ┌──────────┐   ┌──────────┐
          │ Client   │   │ Client   │   │ Client   │
          │ Windows  │   │ Linux    │   │ Mac      │
          └──────────┘   └──────────┘   └──────────┘
```

1.4 Komponen Layanan

Service	Container Name	Base Image	Host Port	Function
Company Profile	company-profile	Debian 12 + Apache	8080	Corporate landing page
WordPress	wordpress	WordPress:latest	8081	Content Management System
Database	wordpress_db	MariaDB:10	Internal	WordPress data persistence
FTPS Server	ftps_server	Debian 12 + vsftpd	21	Secure file transfer

Service	Container Name	Base Image	Host Port	Function
				(Explicit TLS)
SSH Server	ssh_server	Debian 12 + OpenSSH	2222	Remote access + SFTP

1.5 Volume dan Persistensi Data

Volume Name	Mount Path	Content	Purpose
db_data	/var/lib/mysql	WordPress database	Data persistence across container restarts
wp_data	/var/www/html	WordPress core files, plugins, uploads	Website content persistence
ftp_data	/home/ftpuser	FTP user files	Files uploaded via FTPS
ssh_keys	/home/sysadmin/.ssh	SSH authorized_keys	Key-based authentication persistence

---

BAB 2: PENJELASAN TEKNIS PER SERVICE

2.1 Company Profile (Apache HTTP Server)

Fungsi: Menyajikan halaman profil perusahaan yang informatif dan profesional sebagai representasi digital institusi.

Konfigurasi Kunci:

File	Deskripsi
company-profile/Dockerfile	Build image berbasis Debian 12, instalasi Apache2, penyalinan index.html
company-profile/index.html	Halaman HTML dengan CSS terintegrasi yang mencakup visi, misi, layanan, dan kontak

Alur Kerja:

1. Proses docker build membaca instruksi pada Dockerfile
2. Apache2 diinstal ke dalam container
3. File index.html disalin ke direktori /var/www/html/

4. Apache2 dijalankan pada port 80 yang dimapping ke host port 8080

2.2 WordPress dengan MariaDB

Fungsi: Menyediakan platform Content Management System (CMS) yang memungkinkan klien mengelola konten website secara mandiri tanpa intervensi teknis.

Konfigurasi Kunci:

Service	Image	Environment Variables
WordPress	wordpress:latest	WORDPRESS_DB_HOST, WORDPRESS_DB_USER, WORDPRESS_DB_PASSWORD
Database	mariadb:10	MYSQL_ROOT_PASSWORD, MYSQL_DATABASE, MYSQL_USER, MYSQL_PASSWORD

Alur Kerja:

1. Container db (MariaDB) diinisialisasi terlebih dahulu berdasarkan direktif depends_on
2. Container wordpress menunggu hingga database siap menerima koneksi
3. WordPress secara otomatis terhubung ke database menggunakan environment variables yang telah ditentukan
4. Pengguna dapat mengakses http://10.10.10.1:8081 untuk memulai proses instalasi WordPress

2.3 FTPS Server (vsftpd) - Explicit TLS Mode

Fungsi: Menyediakan layanan file transfer dengan enkripsi TLS/SSL menggunakan mode explicit untuk keamanan data saat transmisi.

Rasional Pemilihan Explicit FTPS:

Mode	Port	Karakteristik	Kompatibilitas
Implicit FTPS	990	Enkripsi langsung sejak awal koneksi	Sering mengalami kegagalan handshake
Explicit FTPS	21	Client meminta enkripsi via perintah AUTH TLS	Lebih kompatibel dengan berbagai client

Parameter Konfigurasi Kritis:

Parameter	Value	Penjelasan
listen_port	21	Explicit FTPS menggunakan port standar FTP
ssl_enable	YES	Mengaktifkan fungsionalitas SSL/TLS
force_local_data_ssl	YES	Memaksa enkripsi untuk seluruh transfer data
force_local_logins_ssl	YES	Memaksa enkripsi untuk proses autentikasi
pasv_address	10.10.10.1	Alamat IP server untuk mode passive
seccomp_sandbox	NO	Menonaktifkan seccomp untuk kompatibilitas

Alur Kerja FTPS Explicit:

1. Klien melakukan koneksi ke port 21 (plain FTP)
2. Klien mengirimkan perintah AUTH TLS
3. Server merespons dengan sertifikat SSL/TLS
4. Proses handshake TLS berlangsung
5. Seluruh komunikasi selanjutnya terenkripsi

2.4 SSH Server (OpenSSH)

Fungsi: Menyediakan akses remote yang aman dengan mekanisme autentikasi berbasis kunci publik serta layanan SFTP untuk transfer file.

Parameter Konfigurasi Kunci:

Parameter	Value	Penjelasan
Port	22	Port SSH di dalam container
PasswordAuthentication	yes	Diaktifkan untuk memfasilitasi login pertama
PubkeyAuthentication	yes	Mengizinkan autentikasi berbasis kunci publik
PermitRootLogin	no	Mencegah login sebagai root (best practice keamanan)

Alur Autentikasi SSH Dua Tahap:

Tahap	Metode	Kredensial	Deskripsi
Login Pertama	Password	FirstLoginChangeMe123!	Memudahkan pengguna baru yang belum memiliki SSH key
Setup Key	Manual	Public key pengguna	Pengguna menyalin public key ke

Tahap	Metode	Kredensial	Deskripsi
			~/.ssh/authorized_keys
Login Selanjutnya	Key-based	Private key pengguna	Autentikasi tanpa password setelah key terpasang

Rasional Pendekatan Dua Tahap:

- Mempermudah onboarding pengguna baru yang belum familiar dengan SSH key pair
- Tetap mengutamakan aspek keamanan setelah kunci publik berhasil dipasang
- Mengurangi risiko penggunaan password jangka panjang

BAB 3: PENJELASAN DOCKER COMPOSE

3.1 Network Configuration

text

```
networks:
  internal:
    driver: bridge
```

- Bridge driver: Membangun jaringan virtual internal antar container
- Seluruh container dalam satu network dapat berkomunikasi menggunakan nama service (contoh: db, wordpress)

3.2 Volume Configuration

text

```
volumes:
  db_data:      # Database WordPress
  wp_data:      # File WordPress
  ftp_data:     # File FTP user
  ssh_keys:     # SSH authorized_keys
```

- Volume Docker menyimpan data pada filesystem host
- Data tidak terhapus saat container dihentikan atau dihapus

- Menjamin persistensi data meskipun terjadi restart atau rebuild container

3.3 Service Dependencies

text

```
depends_on:
  - db
```

- WordPress menunggu hingga database siap sebelum memulai proses inisialisasi
- Mencegah error koneksi database pada tahap awal startup

3.4 Environment Variables Management

Seluruh konfigurasi yang bersifat sensitif atau sering berubah disimpan dalam file .env:

Keuntungan	Deskripsi
Separation of Concerns	Konfigurasi tidak di-hardcode dalam docker-compose.yml
Keamanan	Password dan kredensial tidak terekspos di file yang mungkin di-commit ke repositori
Portabilitas	Memudahkan perubahan konfigurasi antar lingkungan (development, staging, production)

BAB 4: PANDUAN INSTALASI

> Prasyarat: Server dengan sistem operasi Debian 12 Bookworm, koneksi internet aktif, dan alamat IP 10.10.10.1

4.1 Struktur Direktori Final

text

```
techinovasi-stack/
├── docker-compose.yml      # Service orchestration
├── .env                    # Environment variables
├── company-profile/
│   ├── Dockerfile          # Apache image build instructions
│   └── index.html          # Corporate profile page
├── ftps/
```

```
│   ├── Dockerfile          # vsftpd image build instructions
│   ├── vsftpd.conf         # FTPS explicit configuration
│   └── entrypoint.sh       # FTPS startup script
├── ssh/
│   ├── Dockerfile          # OpenSSH image build instructions
│   ├── sshd_config         # SSH server configuration
│   └── entrypoint.sh       # SSH startup script
└── data/                   # Persistent data (auto-generated)
    ├── db/
    ├── ftp/
    ├── ssh-keys/
    └── wordpress/
```

4.2 Prosedur Instalasi Bertahap

> Catatan: Seluruh perintah berikut dijalankan pada server 10.10.10.1

STEP 0: Instalasi Docker dan Dependencies

bash

```bash
# Update package index
sudo apt update

# Install dependencies
sudo apt install -y ca-certificates curl gnupg lsb-release tree openssh-client

# Install Docker
curl -fsSL https://get.docker.com -o get-docker.sh
sudo sh get-docker.sh

# Install Docker Compose Plugin
sudo apt install -y docker-compose-plugin

# Add user to docker group
sudo usermod -aG docker $USER
newgrp docker

# Verify installation
docker --version
docker compose version
```

STEP 1: Pembuatan Struktur Direktori

bash

```bash
mkdir -p ~/techinovasi-stack && cd ~/techinovasi-stack
mkdir -p company-profile ftps ssh
mkdir -p data/{db,ftp,ssh-keys,wordpress}
```

STEP 2: Pembuatan File .env

bash

```
nano .env
```

Konten file .env (copy-paste):

env

```
# ============================================
# TECHINOVASI STACK - ENVIRONMENT VARIABLES
# ============================================

# PORT CONFIGURATION
HTTP_PORT=8080
SSH_PORT=2222

# DATABASE CONFIGURATION
MYSQL_ROOT_PASSWORD=rootpass123
MYSQL_DATABASE=wordpress
MYSQL_USER=wpuser
MYSQL_PASSWORD=wppass123

# WORDPRESS CONFIGURATION
WORDPRESS_DB_HOST=db
WORDPRESS_DB_USER=wpuser
WORDPRESS_DB_PASSWORD=wppass123
WORDPRESS_DB_NAME=wordpress

# SSH CONFIGURATION
SSH_USER=sysadmin
SSH_PASSWORD=FirstLoginChangeMe123!
PUBLIC_KEY=""

# FTP CONFIGURATION
FTP_USER=ftpuser
FTP_PASS=ftppass123
```

STEP 3: Pembuatan docker-compose.yml

bash

```
nano docker-compose.yml
```

yaml

```
networks:
  internal:
    driver: bridge

volumes:
  db_data:
  wp_data:
  ftp_data:
  ssh_keys:

services:
  company-profile:
    build: ./company-profile
    container_name: company-profile
    ports:
      - "${HTTP_PORT}:80"
    networks:
      - internal
    restart: unless-stopped

  wordpress:
    image: wordpress:latest
    container_name: wordpress
    ports:
      - "8081:80"
    environment:
      WORDPRESS_DB_HOST: ${WORDPRESS_DB_HOST}
      WORDPRESS_DB_USER: ${WORDPRESS_DB_USER}
      WORDPRESS_DB_PASSWORD: ${WORDPRESS_DB_PASSWORD}
      WORDPRESS_DB_NAME: ${WORDPRESS_DB_NAME}
    volumes:
      - wp_data:/var/www/html
    networks:
      - internal
    depends_on:
      - db
    restart: unless-stopped

  db:
    image: mariadb:10
    container_name: wordpress_db
    environment:
      MYSQL_ROOT_PASSWORD: ${MYSQL_ROOT_PASSWORD}
      MYSQL_DATABASE: ${MYSQL_DATABASE}
      MYSQL_USER: ${MYSQL_USER}
      MYSQL_PASSWORD: ${MYSQL_PASSWORD}
    volumes:
      - db_data:/var/lib/mysql
    networks:
      - internal
    restart: unless-stopped

  ftps:
    build: ./ftps
    container_name: ftps_server
```

```
    ports:
      - "21:21"
      - "40000-40009:40000-40009"
    volumes:
      - ftp_data:/home/ftpuser
    environment:
      FTP_USER: ${FTP_USER}
      FTP_PASS: ${FTP_PASS}
    networks:
      - internal
    restart: unless-stopped

  ssh:
    build: ./ssh
    container_name: ssh_server
    ports:
      - "${SSH_PORT}:22"
    volumes:
      - ssh_keys:/home/${SSH_USER}/.ssh
    environment:
      SSH_USER: ${SSH_USER}
      SSH_PASSWORD: ${SSH_PASSWORD}
      PUBLIC_KEY: ${PUBLIC_KEY}
    networks:
      - internal
    restart: unless-stopped
```

STEP 4: Pembuatan Company Profile

Dockerfile:

bash

```
nano company-profile/Dockerfile
```

dockerfile

```
FROM debian:bookworm-slim
RUN apt-get update && apt-get install -y apache2 && apt-get clean
COPY index.html /var/www/html/index.html
EXPOSE 80
CMD ["apache2ctl", "-D", "FOREGROUND"]
```

index.html: (halaman HTML lengkap dapat dilihat pada lampiran dokumentasi)

STEP 5: Pembuatan FTPS Server

Dockerfile:

bash

```
nano ftps/Dockerfile
```

dockerfile

```
FROM debian:bookworm-slim
RUN apt-get update && apt-get install -y vsftpd openssl && apt-get clean
RUN mkdir -p /etc/vsftpd /var/run/vsftpd/empty /home/ftpuser
RUN openssl req -x509 -nodes -days 365 -newkey rsa:2048 \
    -keyout /etc/vsftpd/vsftpd.key \
    -out /etc/vsftpd/vsftpd.crt \
    -subj "/C=ID/ST=Jakarta/L=Jakarta/O=TechInovasi/CN=10.10.10.1"
COPY vsftpd.conf /etc/vsftpd.conf
EXPOSE 21 40000-40009
COPY entrypoint.sh /entrypoint.sh
RUN chmod +x /entrypoint.sh
ENTRYPOINT ["/entrypoint.sh"]
```

vsftpd.conf:

bash

```
nano ftps/vsftpd.conf
```

conf

```
listen=YES
listen_port=21
anonymous_enable=NO
local_enable=YES
write_enable=YES
local_umask=022
dirmessage_enable=YES
xferlog_enable=YES
connect_from_port_20=YES
secure_chroot_dir=/var/run/vsftpd/empty
pam_service_name=vsftpd
rsa_cert_file=/etc/vsftpd/vsftpd.crt
rsa_private_key_file=/etc/vsftpd/vsftpd.key
ssl_enable=YES
allow_anon_ssl=NO
force_local_data_ssl=YES
force_local_logins_ssl=YES
ssl_tlsv1=YES
ssl_sslv2=NO
ssl_sslv3=NO
require_ssl_reuse=NO
ssl_ciphers=HIGH
```

```
pasv_enable=YES
pasv_min_port=40000
pasv_max_port=40009
pasv_address=10.10.10.1
seccomp_sandbox=NO
```

entrypoint.sh:

bash

```
nano ftps/entrypoint.sh
```

bash

```
#!/bin/bash
if ! id -u $FTP_USER > /dev/null 2>&1; then
    useradd -m -d /home/$FTP_USER -s /bin/bash $FTP_USER
    echo "$FTP_USER:$FTP_PASS" | chpasswd
fi
chown -R $FTP_USER:$FTP_USER /home/$FTP_USER
echo "Starting vsftpd on port 21 (explicit FTPS)..."
vsftpd /etc/vsftpd.conf
```

bash

```
chmod +x ftps/entrypoint.sh
```

STEP 6: Pembuatan SSH Server

Dockerfile:

bash

```
nano ssh/Dockerfile
```

dockerfile

```
FROM debian:bookworm-slim
RUN apt-get update && apt-get install -y openssh-server && apt-get clean
RUN mkdir -p /var/run/sshd /home
COPY sshd_config /etc/ssh/sshd_config
COPY entrypoint.sh /entrypoint.sh
RUN chmod +x /entrypoint.sh
```

```
EXPOSE 22
ENTRYPOINT ["/entrypoint.sh"]
```

sshd_config:

bash

```
nano ssh/sshd_config
```

conf

```
Port 22
Protocol 2
HostKey /etc/ssh/ssh_host_rsa_key
HostKey /etc/ssh/ssh_host_ecdsa_key
HostKey /etc/ssh/ssh_host_ed25519_key
UsePrivilegeSeparation yes
KeyRegenerationInterval 3600
ServerKeyBits 1024
SyslogFacility AUTH
LogLevel INFO
LoginGraceTime 120
PermitRootLogin no
StrictModes yes
RSAAuthentication yes
PubkeyAuthentication yes
AuthorizedKeysFile .ssh/authorized_keys
PasswordAuthentication yes
PermitEmptyPasswords no
ChallengeResponseAuthentication no
UsePAM yes
X11Forwarding yes
PrintMotd no
AcceptEnv LANG LC_*
Subsystem sftp /usr/lib/openssh/sftp-server
```

entrypoint.sh:

bash

```
nano ssh/entrypoint.sh
```

bash

```
#!/bin/bash
ssh-keygen -A
```

```
if ! id -u $SSH_USER > /dev/null 2>&1; then
    useradd -m -d /home/$SSH_USER -s /bin/bash $SSH_USER
    echo "$SSH_USER:$SSH_PASSWORD" | chpasswd
fi
mkdir -p /home/$SSH_USER/.ssh
chmod 700 /home/$SSH_USER/.ssh
if [ ! -z "$PUBLIC_KEY" ] && [ ! -f /home/$SSH_USER/.ssh/authorized_keys ]; then
    echo "$PUBLIC_KEY" > /home/$SSH_USER/.ssh/authorized_keys
    chmod 600 /home/$SSH_USER/.ssh/authorized_keys
    chown -R $SSH_USER:$SSH_USER /home/$SSH_USER/.ssh
    echo "Public key installed."
fi
chown -R $SSH_USER:$SSH_USER /home/$SSH_USER
exec /usr/sbin/sshd -D -e
```

bash

```
chmod +x ssh/entrypoint.sh
```

STEP 7: Generate SSH Key Pair

bash

```
# Generate SSH key untuk client
ssh-keygen -t rsa -b 4096 -f ~/.ssh/techinovasi_client -N ""

# Tampilkan public key (copy output-nya)
cat ~/.ssh/techinovasi_client.pub

# Edit .env dan isi PUBLIC_KEY
nano .env
# Cari baris PUBLIC_KEY="" dan isi dengan key tadi
# Contoh: PUBLIC_KEY="ssh-rsa AAAAB3NzaC1yc2EAAA... client@techinovasi"
```

STEP 8: Menjalankan Seluruh Service

bash

```
cd ~/techinovasi-stack

# Build dan jalankan semua container
docker compose up -d

# Periksa status
docker compose ps

# Periksa log
docker compose logs --tail=30
```

```
# Tunggu inisialisasi database dan WordPress
sleep 60
```

BAB 5: VERIFIKASI DAN PENGUJIAN

5.1 Pengujian dari Server

Test 1: Company Profile

bash

```
curl -s http://localhost:8080 | head -5
```

Expected Result: Menampilkan 5 baris pertama HTML halaman company profile.

Screenshot:

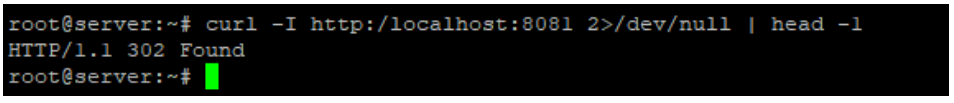
Test 2: WordPress

bash

```
curl -I http://localhost:8081 2>/dev/null | head -1
```

Expected Result: HTTP/1.1 200 OK

Screenshot:



Test 3: FTPS (Explicit TLS)

bash

```
curl -k -u ftpuser:ftppass123 --ssl-reqd ftp://localhost:21 --list-only
```

Expected Result: Daftar file (mungkin kosong) tanpa pesan error.

Screenshot:

```
root@server:~# curl -k -u ftpuser:ftppass123 --ssl-reqd ftp://localhost:21 --list-
only
root@server:~#
```

Test 4: SSH dengan Password

bash

```
ssh -p 2222 sysadmin@localhost "echo 'SSH OK'"
# Password: FirstLoginChangeMe123!
```

Expected Result: SSH OK

Screenshot:

```
C:\Users\nhara>ssh -p 2222 sysadmin@10.10.10.1 "echo 'SSH OK'"
sysadmin@10.10.10.1's password:
SSH OK

C:\Users\nhara>
```

Test 5: SSH dengan Key

bash

```
ssh -p 2222 -i ~/.ssh/techinovasi_client sysadmin@localhost "echo 'SSH KEY OK'"
```

Expected Result: SSH KEY OK (tanpa diminta password)

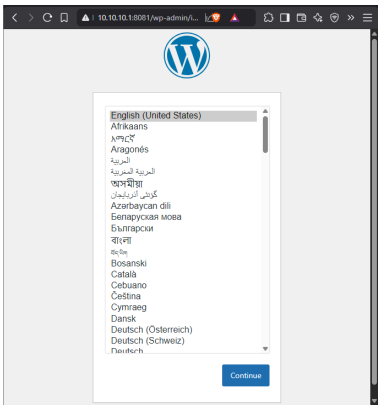
BAB 6: AKSES DARI CLIENT`

6.1 Akses Web

Layanan	URL
Company Profile	http://10.10.10.1:8080
WordPress	http://10.10.10.1:8081

Screenshot:

Screenshot:



6.2 FTPS (File Transfer)

Menggunakan curl (PowerShell/CMD):

powershell

```
curl -k -u ftpuser:ftppass123 --ssl-reqd ftp://10.10.10.1:21 --list-only
```

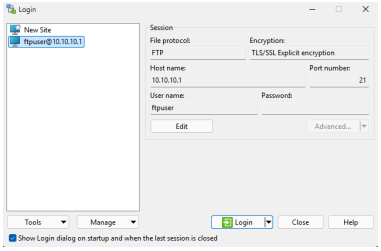
Screenshot:

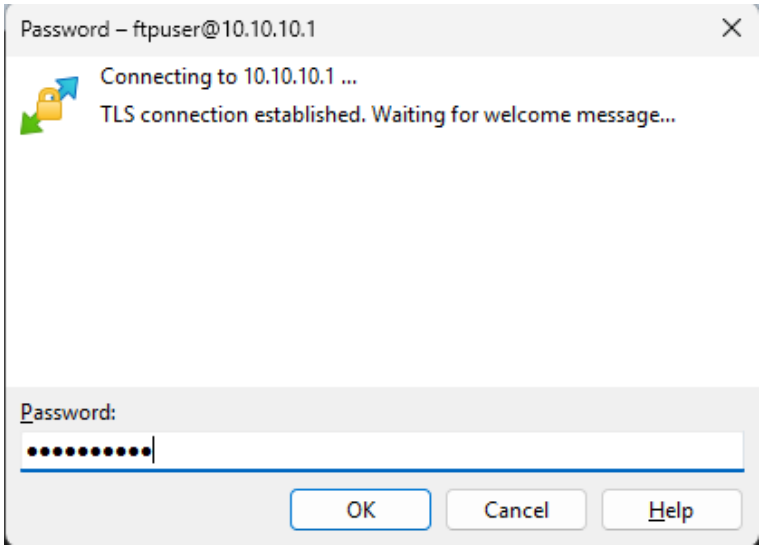
Menggunakan WinSCP:

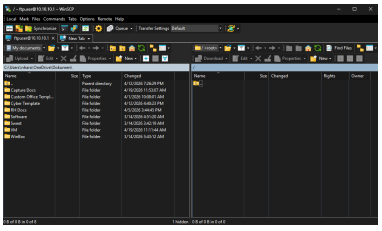
Setting	Value
Protocol	FTP - File Transfer Protocol
Encryption	Require explicit FTP over TLS
Host	10.10.10.1

Setting	Value
Port	21
User	ftpuser
Password	ftppass123

Screenshot:







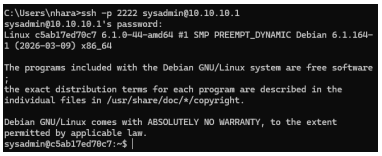
6.3 SSH Remote Access

Login Pertama (Password):

powershell

```
ssh -p 2222 sysadmin@10.10.10.1
# Password: FirstLoginChangeMe123!
```

Screenshot:



Setup SSH Key (Setelah Login):

bash

```
mkdir -p ~/.ssh && chmod 700 ~/.ssh
cat > ~/.ssh/authorized_keys
# Paste public key dari Windows, lalu CTRL+D
chmod 600 ~/.ssh/authorized_keys
exit
```

Login Selanjutnya (Key-based - Tanpa Password):

powershell

```
ssh -p 2222 -i C:\Users\%USERNAME%\.ssh\id_rsa sysadmin@10.10.10.1
```

Step-by-step SSH Testing

Di PowerShell Windows:

powershell

```
# 1. Buat key
ssh-keygen -t rsa -b 4096 -f C:\Users\%USERNAME%\.ssh\id_rsa -N ""

# 2. Lihat public key (COPY hasilnya)
type C:\Users\%USERNAME%\.ssh\id_rsa.pub

# 3. Login pertama (pakai password)
ssh -p 2222 sysadmin@10.10.10.1
# Password: FirstLoginChangeMe123!
```

```
C:\Users\nhara>ssh-keygen -t rsa -b 4096 -f C:\Users\nhara\.ssh\id_rsa -
N ""
Generating public/private rsa key pair.
Your identification has been saved in C:\Users\nhara\.ssh\id_rsa
Your public key has been saved in C:\Users\nhara\.ssh\id_rsa.pub
The key fingerprint is:
SHA256:9VHIGf6GcXnUjVSnuFNvibZjG9bD5+yzsGExpd07E/o nhara@my-workstation
The key's randomart image is:
+---[RSA 4096]----+
|           ..=oo*|
|           .+o.++|
|          . = =..|
|         . . O=oo|
|        S   =*o++|
|            .oB.o|
|             @ B.|
|            + O+=|
|             o E*|
+----[SHA256]-----+

C:\Users\nhara>type C:\Users\nhara\.ssh\id_rsa.pub
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAACAQDO8eqevHEhvWG97VXUcfFA6oQjXJV/awFK
Z95guFttIImgogORh5lus8dOcxQO4Q8EqdCZdkOJzxMR+quRTMg2VmkKMUUVZb5kZRaUxN1d
+cgEyHtH7W2GRyG6LctFC6Zz2kUYVH0liTJfwfxOKK70pRfOXzUN6abXLlfAz6HmQhNhf5kw
JUW3KKC/WGpjKcoKx0RYfk6Xc9o1J3gBS5q/Ih08G/RTswtZ0B+IRsx0nTf7NL1G/qXmhBmE
QBshLHdADWMdvNlWbsGgeBpBNFxjsG1LIYQir/O/sm3w44PiiMdxyBRhmXv57U7X6YnIay67
09t0tb1tJQKThwOxbN7ovCyGFOFblNFJjUGgEZ5dpeq2WVJfINtIxYIzxjH0D8fFd08fKnKC
mocspR95QETTZmCNEdSAZcs3clvoN4E33erEqVkIkxfzHOXH6YKqRZKczekoAkpD/X6XF9W8
ib7NO++1CSiWj+YOlah96zFtPvgvyzICbmeKXJQLjoburCsYFxnxEkSlacLLYq3PXtub2GYM
/Jft38IlFB+m1m4eugIREi24l4+PkhOp6kXJkP4biIsndsAdFBT7KYYpxqVZKkTTHGHKVR15
8aB5wtNLN1KUc3OhOWBwUVZ6ek27tIY0M97Pc+4ul+Nh1Qvpne/xQOrhzT6Ry0wTLPApAgnh
TQ== nhara@my-workstation

C:\Users\nhara>
```

```
PS C:\Users\nhara> ssh -p 2222 sysadmin@10.10.10.1
sysadmin@10.10.10.1's password:
Linux c5ab17ed70c7 6.1.0-44-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.164-
1 (2026-03-09) x86_64

The programs included with the Debian GNU/Linux system are free software
;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Wed Apr 22 06:17:55 2026 from 10.10.10.7
```

Di dalam server (setelah login):

bash

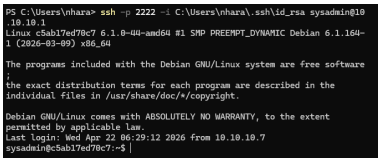
```
mkdir -p ~/.ssh
chmod 700 ~/.ssh
cat > ~/.ssh/authorized_keys
# PASTE public key tadi, lalu ENTER, lalu Ctrl+D
chmod 600 ~/.ssh/authorized_keys
exit
```

```
sysadmin@c5ab17ed70c7:~$ mkdir -p ~/.ssh
sysadmin@c5ab17ed70c7:~$ chmod 700 ~/.ssh
sysadmin@c5ab17ed70c7:~$ cat > ~/.ssh/authorized_keys
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAACAQDO8eqevHEhvWG97VXUcfFA6oQjXJV/awFK
Z95guFttIImgogORh5lus8dOcxQO4Q8EqdCZdkOJzxMR+quRTMg2VmkKMUUVZb5kZRaUxN1d
+cgEyHtH7W2GRyG6LctFC6Zz2kUYVH0liTJfwfxOKK70pRfOXzUN6abXLlfAz6HmQhNhf5kw
JUW3KKC/WGpjKcoKx0RYfk6Xc9o1J3gBS5q/Ih08G/RTswtZ0B+IRsx0nTf7NL1G/qXmhBmE
QBshLHdADWMdvNlWbsGgeBpBNFxjsG1LIYQir/O/sm3w44PiiMdxyBRhmXv57U7X6YnIay67
09t0tb1tJQKThwOxbN7ovCyGFOFblNFJjUGgEZ5dpeq2WVJfINtIxYIzxjH0D8fFd08fKnKC
mocspR95QETTZmCNEdSAZcs3clvoN4E33erEqVkIkxfzHOXH6YKqRZKczekoAkpD/X6XF9W8
ib7NO++1CSiWj+YOlah96zFtPvgvyzICbmeKXJQLjoburCsYFxnxEkSlacLLYq3PXtub2GYM
/Jft38IlFB+m1m4eugIREi24l4+PkhOp6kXJkP4biIsndsAdFBT7KYYpxqVZKkTTHGHKVR15
8aB5wtNLN1KUc3OhOWBwUVZ6ek27tIY0M97Pc+4ul+Nh1Qvpne/xQOrhzT6Ry0wTLPApAgnh
TQ== nhara@my-workstation
sysadmin@c5ab17ed70c7:~$ exit
logout
Connection to 10.10.10.1 closed.
PS C:\Users\nhara>
```

Kembali ke PowerShell Windows:

powershell

```
# 4. Login tanpa password (pakai key)
ssh -p 2222 -i C:\Users\%USERNAME%\.ssh\id_rsa sysadmin@10.10.10.1
```



6.4 SFTP (File Transfer via SSH)

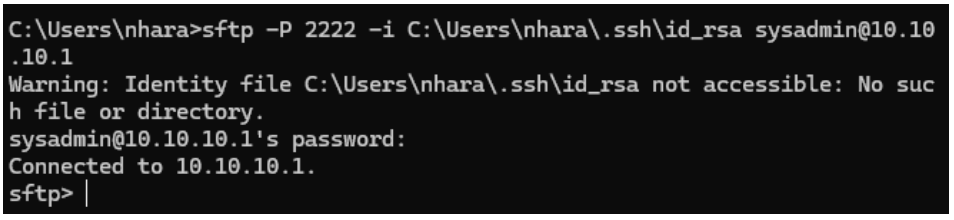
powershell

```
sftp -P 2222 -i C:\Users\%USERNAME%\.ssh\id_rsa sysadmin@10.10.10.1
```

Perintah di dalam SFTP:

Command	Fungsi
ls	Menampilkan daftar file di server
put file.txt	Mengunggah file dari lokal ke server
get file.txt	Mengunduh file dari server ke lokal
exit	Keluar dari session SFTP

Screenshot:



BAB 7: MANAJEMEN OPERASIONAL

7.1 Perintah Docker Compose

bash

```
cd ~/techinovasi-stack

# Mengaktifkan semua service
docker compose up -d

# Menghentikan semua service
docker compose down

# Restart service tertentu
docker compose restart ftps
docker compose restart ssh
docker compose restart wordpress

# Melihat status service
docker compose ps

# Melihat log secara realtime
docker compose logs -f

# Melihat log service tertentu
docker compose logs -f ftps
```

7.2 Prosedur Backup dan Restore

Backup seluruh data:

bash

```
docker compose down
tar -czf backup-techinovasi-$(date +%Y%m%d).tar.gz ~/techinovasi-stack/
docker compose up -d
```

Restore data:

bash

```
docker compose down
tar -xzf backup-techinovasi-YYYYMMDD.tar.gz -C /
docker compose up -d
```

7.3 Prosedur Update Service

Update WordPress:

bash

```
docker compose pull wordpress
docker compose up -d --build wordpress
```

Update custom service (Company Profile/FTPS/SSH):

bash

```
docker compose build --no-cache ftps
docker compose up -d ftps
```

BAB 8: TROUBLESHOOTING

Masalah	Kemungkinan Penyebab	Solusi
Connection refused pada FTPS	Service FTPS tidak berjalan	docker compose logs ftps
wrong version number pada FTPS	Menggunakan implicit mode	Gunakan --ssl-reqd (explicit mode)
SSH masih meminta password	Public key tidak tersalin dengan benar	cat ~/.ssh/authorized_keys di server
WordPress error koneksi database	Database belum siap	Tunggu 30 detik, docker compose restart wordpress
Port already in use	Konflik port dengan service lain	Ubah port di file .env
Permission denied (docker)	User tidak memiliki akses docker group	sudo usermod -aG docker $USER lalu logout/login

---

BAB 9: SNAPSHOT DAN ROLLBACK STRATEGY

Snapshot	Waktu Eksekusi	Nama Snapshot	Tujuan
1	Setelah instalasi Docker	1-docker-installed	Baseline lingkungan bersih
2	Sebelum docker compose up -d pertama	2-before-first-up	Sebelum eksperimen konfigurasi
3	Setelah semua service berjalan	3-all-running	Point of success

Snapshot	Waktu Eksekusi	Nama Snapshot	Tujuan
4	Setelah SSH key berhasil dipasang	4-ssh-key-working	Keamanan terkonfigurasi
5	Final sebelum serah terima ke klien	5-ready-for-client	Produk final

---

BAB 10: REFERENSI KREDENSIAL

10.1 Port Mapping

Service	Container Port	Host Port
Company Profile	80	8080
WordPress	80	8081
FTPS	21	21
FTPS Passive Range	40000-40009	40000-40009
SSH/SFTP	22	2222

10.2 Kredensial Default

Service	Username	Password / Key
FTPS	ftpuser	ftppass123
SSH (first login)	sysadmin	FirstLoginChangeMe123!
SSH (subsequent)	sysadmin	SSH private key
MySQL (root)	root	rootpass123
MySQL (WordPress)	wpuser	wppass123

> PENTING: Seluruh password default WAJIB diganti sebelum deployment ke lingkungan production untuk memenuhi standar keamanan.

BAB 11: KESIMPULAN

TechInovasi Stack berhasil menyediakan solusi infrastruktur dengan spesifikasi sebagai berikut:

Aspek	Status	Keterangan
Automasi Total	Terpenuhi	Satu perintah (docker compose up -d) untuk

Aspek	Status	Keterangan
		seluruh service
Keamanan Berlapis	Terpenuhi	FTPS (TLS) + SSH key-based authentication
Data Persisten	Terpenuhi	Volume Docker untuk seluruh data kritis
Cross-Platform Access	Terpenuhi	Kompatibel dengan Windows, Linux, dan Mac
Production Ready	Terpenuhi	Seluruh konfigurasi telah melalui proses pengujian

---

LAMPIRAN: FILE KONFIGURASI LENGKAP

Seluruh file konfigurasi telah dicantumkan pada BAB 4 (Panduan Instalasi) STEP 2 hingga STEP 6.

Ringkasan file yang perlu dibuat:

File Path	Fungsi
~/techinovasi-stack/.env	Environment variables
~/techinovasi-stack/docker-compose.yml	Service orchestration
~/techinovasi-stack/company-profile/Dockerfile	Apache image build
~/techinovasi-stack/company-profile/index.html	Corporate web page
~/techinovasi-stack/ftps/Dockerfile	vsftpd image build
~/techinovasi-stack/ftps/vsftpd.conf	FTPS configuration
~/techinovasi-stack/ftps/entrypoint.sh	FTPS startup script
~/techinovasi-stack/ssh/Dockerfile	OpenSSH image build
~/techinovasi-stack/ssh/sshd_config	SSH configuration
~/techinovasi-stack/ssh/entrypoint.sh	SSH startup script

Dokumentasi ini siap digunakan untuk keperluan serah terima kepada klien maupun sebagai referensi internal tim teknis.

TechInovasi Stack - Professional IT Infrastructure Automation